Individual Coursework Submission Form

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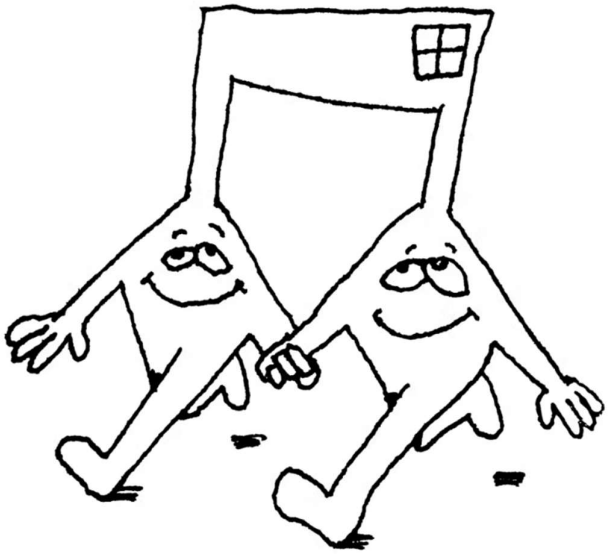
THIS IS DICE



Backstage Intelligence: Making Every Seat Count at DICE

A Strategic Business Analytics Report for Optimising Demand, Attendance, and Engagement at DICE

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“DICE’s mission is to get people out more — by making live event discovery easy, trustworthy, and personal.”

1 About DICE

1.1 Background Information

DICE is a London-based ticketing platform operating in the live entertainment industry, known for its mobile-first, fan-friendly approach to event discovery and booking. Launched in 2014, the company has grown into a global player, offering curated access to concerts, club nights, festivals, and comedy shows across key cities including London, New York, Los Angeles, Berlin, and Milan.

At the heart of DICE’s business model is a commitment to transparency and trust. The platform removes hidden fees, prevents ticket resale through a locked system, and uses waitlists to ensure fair access when events sell out. Revenue is primarily generated from ticketing service fees and promoter partnerships, while value is delivered through seamless user experiences, personalized event discovery, and strong artist-fan connections.

Despite securing over $100M in funding and acquiring Boiler Room to expand its cultural footprint, DICE operates in a highly competitive and fast-evolving market dominated by players like Ticketmaster and Eventbrite. As demand for live experiences rebounds post-COVID, challenges such as last-minute bookings, no-shows, and discovery friction continue to impact growth, operational efficiency, and user retention.

This report identifies and addresses three strategic problems where DICE is underperforming due to limited or suboptimal use of data analytics. Each section proposes targeted, data-driven solutions aligned with business objectives, supported by implementation plans and frameworks to guide decision-making at the executive level.

1.2 Business Model Canvas

DICE is a mobile-first ticketing platform designed to serve two core customer segments: fans seeking trusted, curated access to live events, and promoters, artists, and venues looking for ethical, data-driven ticketing solutions. Its core value proposition lies in removing traditional ticketing friction—no hidden fees, no reselling—and offering fair access through waitlists, alongside a personalized discovery experience that encourages greater event attendance.

The platform’s main revenue stream comes from service fees on ticket sales, paid by users. DICE also charges partners for platform access and tools that enhance audience targeting, promotions, and performance insights. With a growing focus on livestreaming and brand partnerships, DICE is expanding its monetization model beyond traditional ticketing.

User engagement is driven through the DICE app and website, with features like tailored event feeds, smart notifications, and waitlists that support long-term retention. These channels also underpin community building and data collection. Internally, operations rely on proprietary technologies—

recommendation algorithms, behavioural data models, and fraud prevention tools—supported by teams in engineering, product, and marketing.

Key partners include independent venues, promoters, artists, and platforms like Spotify, which enhances user profiling and recommendations. The company's cost structure is primarily driven by technology development, expansion, and relationship management. Centred around the mission to "get people out more," DICE positions itself as a transparent, modern alternative in a live entertainment landscape dominated by legacy ticketing giants.

2 Forecast First: Embedding Predictive Demand Planning into Event Operations

2.1 Problem Statement: Unpredictable Demand and Last-Minute Bookings

One of the most pressing challenges faced by DICE is the unpredictable nature of ticket demand, particularly due to the high volume of last-minute bookings. Internal partner reports suggest that approximately 40% of tickets are sold within the final 48 hours leading up to an event. This pattern makes it difficult for promoters and internal teams to anticipate audience size with confidence, creating operational and financial inefficiencies across the board.

When demand cannot be forecasted accurately, venues risk being either underprepared or overstaffed, which directly impacts both the cost structure and the audience experience. Marketing campaigns are often deployed too late or too broadly, resulting in poor return on investment and lower ticket conversion rates. Additionally, DICE misses the opportunity to leverage time-sensitive strategies such as early bird discounts, surge pricing, or targeted promotions during key demand windows. This unpredictability weakens DICE's value proposition for both fans and partners and constrains its ability to scale effectively in new cities or optimize event economics.

2.2 Analytic Strategy

To address this issue, it is recommended that DICE implement a predictive demand forecasting model to anticipate event-level ticket sales with greater lead time and accuracy. This would shift the business from reactive event management to proactive planning, directly supporting operational efficiency and enhancing promoter trust.

The model would integrate three analytical layers. First, time series forecasting methods such as Prophet or ARIMA would detect patterns in historical ticket sales across similar events. Second, event metadata clustering would group events by shared characteristics like genre, venue size, artist popularity, and day of the week. Third, user behavioural segmentation would factor in real-time indicators such as click-through rates, wish lists, and in-app engagement at a city level. Combined, these layers would produce demand curves for each event well ahead of peak booking periods.

This solution aligns with DICE's strategic goals by enabling more efficient resource planning, smarter marketing activation, and earlier waitlist management. It also supports market expansion by reducing uncertainty for promoters in new cities.

The implementation is supported by multiple frameworks. Using Digital Product Management, the forecasting tool becomes part of the promoter's planning interface. The Agile Scrum approach allows for rapid iteration and model refinement. Finally, Design Thinking ensures the solution is co-designed with promoter feedback, leading to intuitive features such as early-warning alerts and forecast confidence scores.

2.3 Action Plan

The project would begin by defining forecast accuracy targets and gathering historical ticketing and user behaviour data. In the second phase, the data science team would develop and validate forecasting models by event type and city. These models would then be integrated into a dashboard for use by promoters and internal stakeholders.

A pilot rollout in selected cities would follow, allowing the product team to refine the interface based on user feedback. Once operational, the model would be retrained weekly using live sales and behavioural data. Feedback loops from actual event outcomes would be used to improve forecast accuracy over time. Forecasting performance would be measured by tracking deviations from actual attendance, with an initial target of maintaining predictions within ±15% accuracy for the majority of events.

3 Smart Seats: Reducing No-Shows with Predictive Attendance Intelligence

3.1 Problem Statement: High No-Show Rates Despite "Sold-Out" Events

A significant operational issue for DICE is the disconnect between ticket sales and actual attendance [Appendix C: Figure 1]. Despite events being marked as "sold out," a substantial proportion of fans fail to attend particularly when tickets are free, discounted, or RSVP-based. According to DICE's internal assessments and feedback from promoters, no-show rates in these cases can exceed 30%.

This misalignment creates visible empty spaces at live events, negatively impacting the energy of the experience for artists and attendees alike. More importantly, it damages the trust that promoters and venues place in the platform's ability to deliver reliable turnout. DICE's existing waitlist system helps to redistribute tickets, but currently lacks predictive capabilities to proactively manage drop-offs in real time. As a result, tickets remain unused, revenue opportunities are lost, and customer satisfaction is compromised.

3.2 Analytic Strategy

To improve attendance reliability, it is recommended that DICE implement a No-Show Probability Prediction Model. This model would identify, in advance, the likelihood that a given ticket holder will not attend a particular event. The approach would use classification techniques such as logistic regression or random forest, informed by behavioural and transactional indicators.

Key variables would include ticket type (paid, RSVP, or promotional), the timing of purchase, historical attendance records, and engagement signals like reminder views and prior check-in history. Once probabilities are assigned, the system can automatically initiate targeted nudges, such as reminder notifications or exclusivity messages to users flagged as high-risk. If the likelihood of no-shows surpasses a threshold, the model can trigger waitlist promotions or re-release inventory to other interested users in real time.

This solution is strategically aligned with DICE's objective to optimize event turnout and enhance partner confidence. It transforms the waitlist system from a passive feature into a dynamic reallocation engine, improving operational efficiency and maximizing the value of each ticket issued.

The initiative draws on multiple strategic frameworks. Through Digital Transformation, the waitlist becomes a smart, self-adjusting tool embedded in the core ticketing workflow. Applying Value Proposition Design, the system delivers a stronger fan experience, creating moments like last-minute ticket wins—while restoring confidence among promoters. Finally, using Project Management

principles ensure cross-functional coordination between data, product, and marketing teams for effective deployment.

3.3 Action Plan

The project will begin by labelling and segmenting historical attendance data to train the prediction model. In the second phase, the data science team will validate and refine the model's accuracy using upcoming event data. Once proven, the system will be integrated into the ticketing platform to automate reminders and waitlist adjustments. Ongoing monitoring will assess attendance improvements and adjust model thresholds as needed.

The solution will be overseen by the data science team, with support from marketing for messaging strategies and the product team for system integration. Success will be measured by a targeted reduction in no-show rates, ideally achieving a 20% improvement across RSVP-based events within the first three months of deployment.

4 For You, Not for All: Personalizing Discovery with Data-Driven Precision

4.1 Problem Statement: Underperforming Personalization in Event Discovery

DICE's growth relies heavily on its ability to match fans with events they genuinely want to attend. However, user feedback indicates that the platform's current recommendation system is not consistently delivering relevant suggestions, particularly for new users or those in recently launched markets like New York or Berlin. As a result, users find it difficult to discover events aligned with their interests, leading to lower engagement, reduced app retention, and missed connections between artists and their target audiences. [Appendix C- Figure 3]

Public reviews on the App Store and Reddit forums often mention irrelevant event suggestions and limited variety in notification content. In an industry increasingly driven by personalization, DICE's underutilization of its rich behavioural data weakens its competitive edge, especially when compared to music discovery platforms like Spotify, which excel at using data to enhance user relevance and habit formation.

4.2 Analytic Strategy

To address this, DICE should upgrade its recommendation engine by deploying a hybrid personalization model. This would combine collaborative filtering techniques (such as matrix factorization and KNN) with natural language processing (NLP) applied to artist bios, event descriptions, and external listening data. For new users, where historical engagement is minimal, cold-start optimization techniques such as interest surveys and demographic clustering can be introduced during onboarding.

The system would dynamically generate a "For You" feed tailored to each user's preferences and behaviour. Input data would include past bookings, likes, and waitlist activity, as well as interactions by users with similar profiles. External integrations, such as Spotify listening history or trending artist data, would further enrich recommendations.

This solution directly supports DICE's strategic goal of improving user retention and maximizing artist reach. Stronger personalization increases time spent in the app, improves the discovery-to-purchase conversion rate, and strengthens DICE's position as a go-to platform for curated event experiences.

The initiative draws on multiple frameworks. Design Thinking informs the onboarding flow and recommendation UI by focusing on first-time user confusion and decision fatigue. Through the Business Model Canvas, personalization is framed as a core activity that drives customer value. Finally, an Agile MVP approach allows DICE to test the improved engine in a few high-priority cities, gather feedback, and iterate quickly before scaling.

4.3 Action Plan

The initial phase would involve consolidating behavioural data with artist metadata and external signals. A short onboarding survey would also be introduced to collect preference data for new users. In the second phase, the data science team would train a hybrid model and evaluate performance across recommendation accuracy and click-through rate benchmarks.

In month three, the product and UX teams would run A/B testing on the redesigned homepage layout, comparing the new personalized feed against the current static discovery section. Based on performance insights, further iterations would follow. The model would then be continuously refined using real-time engagement data and user feedback.

The initiative would be led by the data science team, in collaboration with product, engineering, and customer insights. Success metrics would include a 15% improvement in click-through rate and a measurable reduction in first-month churn among new users.

5 Conclusion

This report identified three key challenges impacting DICE's performance: unpredictable demand, high no-show rates, and weak personalization in event discovery. Each issue limits DICE's ability to scale, retain users, and deliver consistent value to its partners.

To address these gaps, the report proposed three data-driven solutions: a predictive demand forecasting model to improve operational planning, a no-show probability engine to enhance attendance reliability, and a hybrid recommendation system to boost user engagement. Each initiative is directly aligned with DICE's strategic goals, that is, improving event performance, deepening user retention, and increasing promoter trust.

Beyond the technical recommendations, these solutions represent a broader shift toward embedding analytics into DICE's core business processes. Implemented through agile, cross-functional collaboration and guided by frameworks like Design Thinking and Digital Product Management, they position DICE to deliver smarter, more scalable experiences and stay true to its mission of getting more people out to the events they love.

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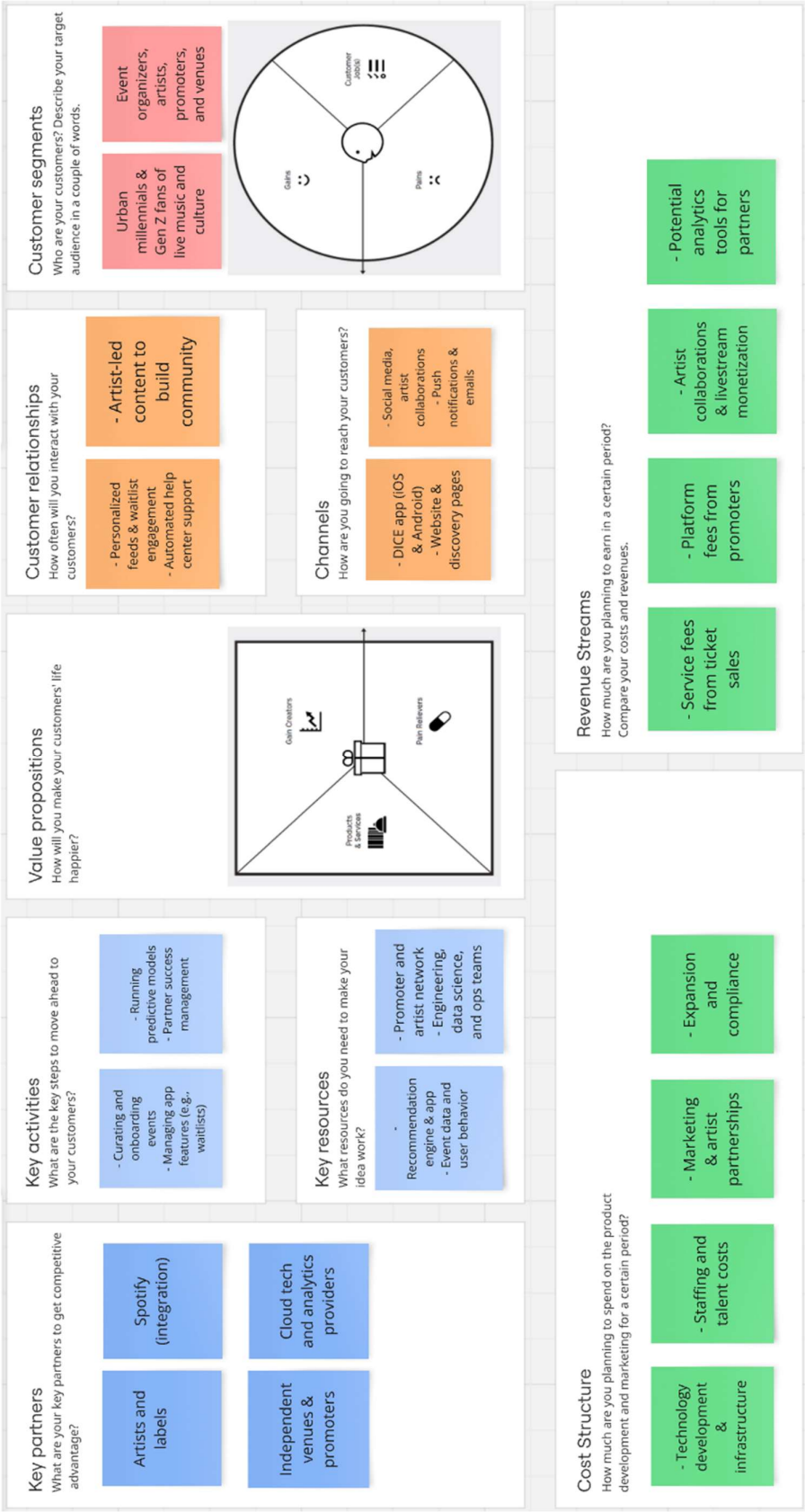
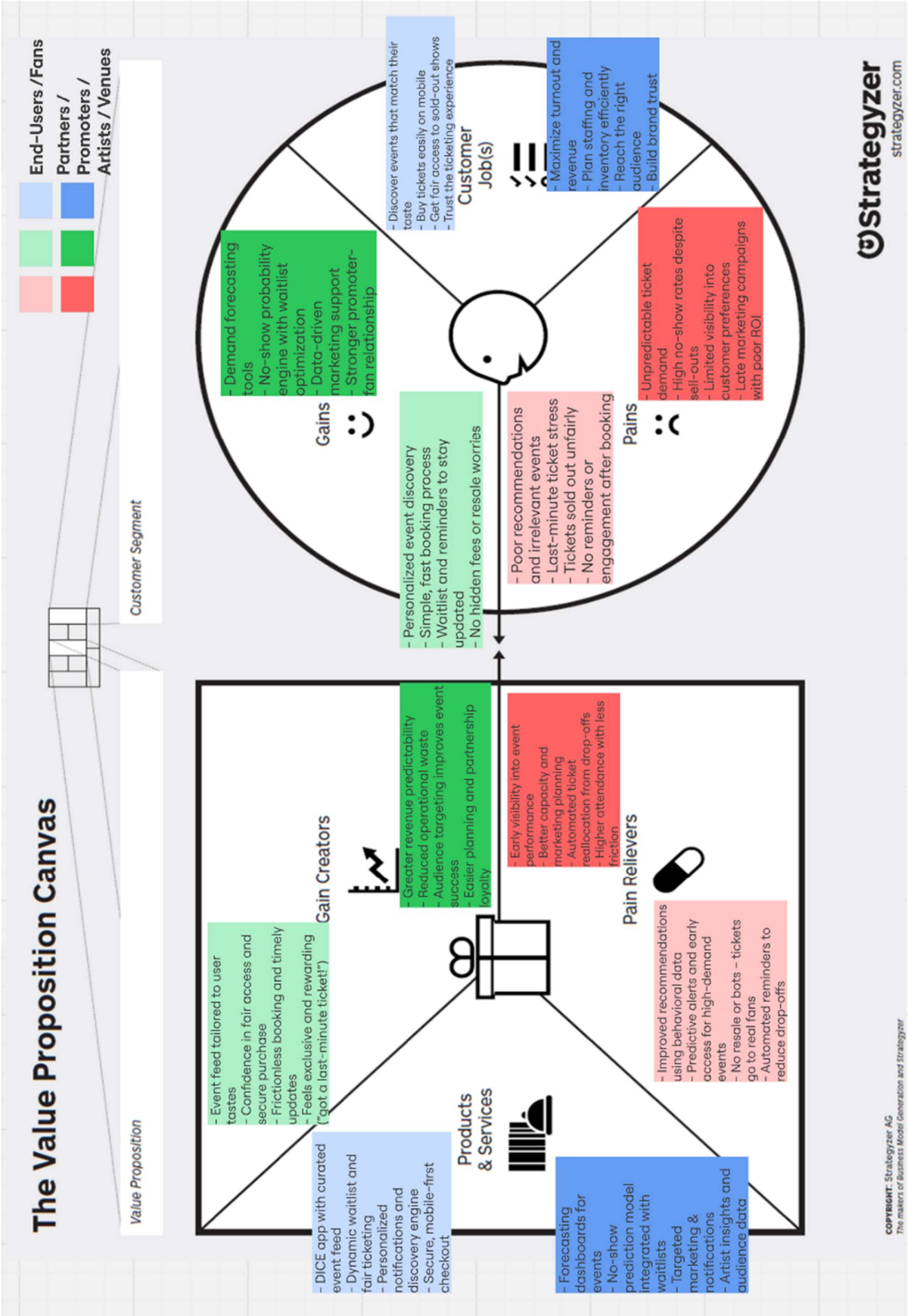
Appendix

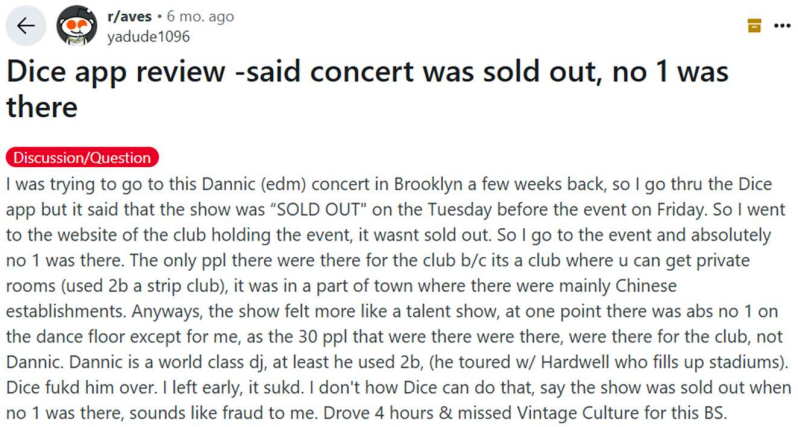
Appendix A: Business model canvas



Appendix B: Value Proposition Canvas



Appendix D: Evidences



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yadude1096

Dice app review -said concert was sold out, no 1 was there

Discussion/Question

I was trying to go to this Dannic (edm) concert in Brooklyn a few weeks back, so I go thru the Dice app but it said that the show was "SOLD OUT" on the Tuesday before the event on Friday. So I went to the website of the club holding the event, it wasnt sold out. So I go to the event and absolutely no 1 was there. The only ppl there were there for the club b/c its a club where u can get private rooms (used 2b a strip club), it was in a part of town where there were mainly Chinese establishments. Anyways, the show felt more like a talent show, at one point there was abs no 1 on the dance floor except for me, as the 30 ppl that were there were there, were there for the club, not Dannic. Dannic is a world class dj, at least he used 2b, (he toured w/ Hardwell who fills up stadiums). Dice fukd him over. I left early, it sukd. I don't how Dice can do that, say the show was sold out when no 1 was there, sounds like fraud to me. Drove 4 hours & missed Vintage Culture for this BS.

Figure 1: A Reddit user shared their experience: "Dice app review - said concert was sold out, no 1 was there."

Mobile ticket app Dice has introduced what is believed to be a world first in concert ticketing: refunds for those who can no longer attend.

Announcing the news yesterday, Dice founder Phil Hutcheon said refunds are initially only available for sold-out events (although, he adds, "over 80%" of Dice-ticketed shows sell out) but that the new initiative should benefit buyers and sellers alike: the former "no longer have to scour secondary ticket websites" for sold-out shows, instead adding themselves to a waiting list for returned tickets, and the latter can rest safe in the knowledge they're not supporting secondary platforms frequented by "speculators and brokers who hoover up [...] tickets and immediately resell them".

Figure 2: An article from IQ Magazine discusses DICE's efforts: "Dice introduces ticket refunds... Those unable to attend sold-out shows can now return their tickets for a no-fuss refund."

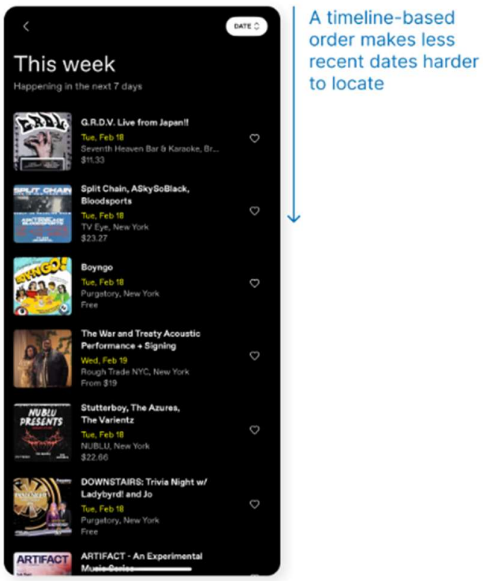

Figure 3: Event Discovery View on DICE App
This screenshot demonstrates a timeline-based layout lacking personalization. Users are shown all upcoming events in a static format, making relevant events harder to find—especially in large or unfamiliar cities.

Appendix D: Action Plans

Table 1: Forecast First: Embedding Predictive Demand Planning into Event Operations

Phase	***Steps***	***Who's Responsible***
Month 1	Define KPIs (forecast error, lead time gain). Collect sales and behaviour data from DICE systems.	**Data & Analytics Team** (lead), Product Team, Event Partnerships
Month 2	Train forecasting models on historical data segmented by city, genre, and event type.	**Data Science Team** (lead), Event Ops Team for validation
Month 3	Develop interactive dashboards for DICE's internal team and key partners.	**Product Team** (lead), UI/UX Designers, Marketing
Ongoing	Weekly model updates with live data + feedback loop from event outcomes.	**Data Engineering + Analytics** (lead), Event Managers

Table 2: Smart Seats: Reducing No-Shows with Predictive Attendance Intelligence

Phase	***Steps***	***Who's Responsible***
Month 1	Segment past events by no-show rate. Label training data using check-in records and user behaviour logs.	**Data Science Team** (lead), Customer Experience Team
Month 2	Train classification model and test predictive accuracy on upcoming events.	**Data Science Team** (lead), Product Team
Month 3	Integrate model into ticketing system. Auto-trigger reminders and waitlist reshuffling.	**Engineering Team** (lead), CRM/Marketing Team
Ongoing	Monitor attendance uplift and optimize thresholds for action.	**Analytics Team** (lead), Event Operations, Customer Insights

Table 3: For You, Not for All: Personalizing Discovery with Data-Driven Precision

Phase	***Steps***	***Who's Responsible***
Month 1	Merge in-app behaviour logs with Spotify and artist data. Design user onboarding survey for preference collection.	**Data Engineering** (lead), Product, Partnerships (for Spotify), Design Team
Month 2	Train hybrid recommendation engine using collaborative + content-based filtering.	**Data Science Team** (lead), Front-End Engineering
Month 3	A/B test new homepage layout with personalized feed vs generic discovery. Track click-through and booking rates.	**Product Team** (lead), UX Researchers, Analytics Team
Ongoing	Continuously learn from user interactions to fine-tune models.	**Data Science** (lead), Marketing, Customer Support for qualitative feedback

Appendix E: Pros and Cons of Implementing Data Analytics at DICE

Pros	Cons
Improves demand forecasting, allowing DICE and partners to plan ahead and reduce reliance on last-minute bookings	Requires a **high volume of clean, historical data**, which may not exist for new cities or first-time events
Reduces no-show rates by identifying at-risk users and reallocating tickets via predictive models	**Integration complexity** with existing ticketing and waitlist systems can delay implementation
Enhances personalization in event discovery, increasing user engagement, retention, and ticket conversion	Cold-start problems may affect personalization accuracy for **new users or artists with limited data**
Enables data-driven decision-making across product, marketing, and operations	Initial implementation demands **technical expertise, training, and cross-functional coordination**
Strengthens DICE's value proposition to promoters and artists by delivering more reliable turnout and targeted audience insights	May raise **data privacy and compliance** concerns (e.g., GDPR) when handling behavioural or third-party data
Positions DICE as a modern, intelligence-led platform, differentiating it from traditional ticketing competitors	**Cost of investment** in infrastructure, tooling, and continuous model retraining can be substantial